Perceptron Hackathon 2026 | Problem 4 | Theme 3: Markets & Trading Intelligence

Team: Attention Is All You Need | **Team ID:** TM - 26AO64E9F9D | **Members:** Arnav Garg & Aditya Viraj

Problem and motivation

Short-horizon order-book forecasts assume, often implicitly, that the relationship between current book conditions and future price movement remains stable. In practice, liquidity, spreads, volatility and order flow change, and a model that performs well in one period may become unreliable in another. This is difficult to evaluate when directional moves are rare. With our 5-basis-point movement buffer, 95.46% of the cleaned observations are labelled neutral. A model can therefore achieve high ordinary accuracy by mostly predicting no movement while missing less frequent directional events.

We approach Problem 4 as both a forecasting and a reliability problem: identify when a change is associated with deteriorating forecast quality, respond using only information already available, and make uncertainty visible.

Data and prediction task

We use NASDAQ limit order book and message data from LOBSTER. Our prepared dataset contains 624,040 chronological observations with ten levels of book depth. We construct multi-level depth imbalance, bid–ask spread and rolling log-return volatility features. The target is the future mid-price movement over a ten-observation horizon: up above +5 basis points, down below -5 basis points, and neutral within the buffer. The cleaned-data proportions are 2.18% up, 2.36% down and 95.46% neutral.

Proposed system

Our approach has four connected components.

First, a regularised Random Forest provides the baseline forecast. Class weights address the rare directional labels, while constrained tree growth limits overfitting. We evaluate its probabilities using out-of-sample log loss and class-sensitive metrics; regularisation is not assumed to guarantee calibration.

Second, we monitor prequential log loss. Each prediction is scored only when its ten-observation-ahead label becomes available. ADWIN monitors this sequence and signals a change when the averages of subwindows differ sufficiently. We treat an alarm as evidence of a change in predictive performance, not proof of a distinct economic regime. Because labels arrive late, this signal is necessarily reactive.

Third, after a qualifying alarm, ShiftGuard is designed to retrain the Random Forest using a recent window of already labelled observations. This avoids learning from unavailable

outcomes and lets us test whether detector-triggered retraining is more useful than a frozen model or fixed retraining schedule.

Fourth, a reliability gate combines recent loss evidence with current volatility. It can pass a forecast through, flag it as degraded, or abstain. Volatility provides a contemporaneous market-condition signal, while the loss monitor provides delayed evidence about predictive performance. Neither is treated as a calibrated probability of correctness; we will evaluate the gate through selective performance and coverage.

Minimum viable product and evaluation

The MVP will forecast three-class mid-price movement, monitor prequential loss with ADWIN, and demonstrate the model's response to detected performance changes. We will use chronological walk-forward evaluation, respect the ten-observation label delay, and purge training observations whose forward labels overlap the test boundary. Model and detector settings will be selected on validation data, leaving the final test period untouched.

We will compare: (1) a frozen Random Forest; (2) periodic retraining, with its schedule selected on validation data; (3) ADWIN-triggered retraining without the gate; and (4) full ShiftGuard. The periodic baseline will use a comparable retraining budget, testing whether the detector adds value beyond retraining itself. Metrics will include log loss, balanced accuracy, macro-F1, per-class precision and recall, alarm frequency, post-alarm performance and retraining cost. We will examine alarms alongside spread and volatility, and measure the gate's log loss at different levels of forecast coverage. Detection delay against known change points will be measured on separately identified or controlled changes, not by treating unverified historical alarms as ground truth.

Novelty, feasibility and limitations

Our Approach combines performance-based change monitoring, delayed-label-aware retraining and an explicit reliability decision. Rather than flagging input changes alone, it asks whether forecast errors have changed. Rather than updating continuously, it retrains after a qualifying signal using matured labels. The gate makes abstention an explicit output instead of assuming every forecast should be acted upon.

The LOBSTER sample provides a practical setting for studying intraday order-book changes, but may not contain major macroeconomic events or represent all market regimes. We will describe observed shifts conservatively and use controlled shifts only as stress tests. ADWIN may alarm too often or too late, directional labels are sparse, and gains over a strong prior baseline may be modest; our evaluation is designed to test these risks rather than assume success.

Our team combines advanced quantitative finance and statistical understanding with intermediate Python and machine-learning skills. The MVP uses established Python tools and is designed for a laptop. We aim to complete the core pipeline and baseline before the November 10 shortlist, the walk-forward detector and retraining experiment by the December 1 midway report, and the reliability gate and demonstration before the December 15-18

finals. Regime-specific experts and detailed latency benchmarking are stretch goals after the core system is validated.